

Group Start MAPS Template (Instrumentation)

The Group Start MAPS template is designed to control a process.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in two models – advanced and standard, which provide respectively lower levels of control over the group.

This document describes the Advanced Group Start MAPS Template and also defines the level of functionality provided by the standard model. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the group being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced Group Start Functional Philosophy

The group start outputs are enabled by the operator, once requests are received.

The Start, Stop and Reset requests are set by the operator from the faceplate.

The faceplate will then display the current status (Starting, Stopping, Stopped or Running) of the group.

Typical Faceplate Graphics for the Advanced Group Start MAPS Template

Group Start Advanced	Group Start Standard	
		
	Start Request	
	Stop Request	
	Reset Request	
	Group Name	

Graphics Legend:		
Start Button	Stop Button	Group status
Light Green	Dark Red	Stopped
Light Green	Medium Red	Stopping
Medium Green	Light Red	Starting
Dark Green	Light Red	Running

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Start Pulse		x	x	
Bit 1	Stop Pulse		x	x	
Bit 2	Reset Pulse		x		
Bit 3	Spare				
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Stopped		x	x	
Bit 1	Stopping		x	x	
Bit 2	Starting		x	x	
Bit 3	Running		x	x	
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

Resources	Advanced	Standard	Basic
Digital Inputs	0	0	
Digital Outputs	0	0	
Analog Inputs	0	0	
Analog Outputs	0	0	
PLC Memory Steps	92	89	
System Words Used	0	0	
System Bits Used	60	59	
SCADA SCAN Points	2	2	